0101026/27 2&1 B.B.E.K.O
October 2026
MATHEMATICS 2&1 **2 &1**
Essay & Objective
2 hours

Name:.....

Index Number:.....

BEST BRAIN EXAMINATIONS KONSORTIUM
GHANA

Special Private Mock Examinations for BECE Candidates

October 2026 **MATHEMATICS 2 & 1** 2 hours

Do **not** *open this booklet until you are told to do so. While you are waiting, read and observe the following instructions carefully. Write your* **name** *and* **index number** *in* **ink** *in the spaces provided above.*

This booklet consists of **two** *papers. Answer Paper* **2** *which comes first, in your answer booklet and Paper* **1** *on your Objective Test answer sheet. Paper* **2** *will last* **1** *hour after which the answer booklet will be collected. Do* **not** *start Paper* **1** *until you are told to do so. Paper* **1** *will last* **1** hour.

The use of calculators is **not** *allowed.*

Acadmicprep.com
Learn • Practice • Excel

© 2026 Best Brain Examinations Konsortium
(020-7732900 / 0249273049)

CamScanner

PAPER 2

1 hour

Essay

Answer **four** *questions* **only.**
All questions carry equal marks.

All *workings must be clearly shown. Marks will* **not** *be awarded for* **correct** *answers without corresponding working.*

1. (a) Convert the following and express your answers in standard form:
(i) 31 days into minutes;
(ii) 6 hours into seconds.

(b) Fred earns GH₵ 2500 a month after tax and Eben earns three times as much. How much is their total income after five years if there are no increases in their earnings?

(c) The following set of data were collected from an interview:

2	**3**	**(4m – 3)**
3m + 1	**11**	**13**

If the median in ascending order is 6, find the value of m.

(d) The following are the marks obtained by 20 students in a class examination.

68	42	63	57	66
54	72	58	57	73
61	54	65	45	38
40	57	58	56	48

Acadmicprep.com
Learn • Practice • Excel

(i) Arrange the marks above on a stem – and – leaf plot in ascending order
(ii) What is the mode?
(iii) Determine the median.

2. (a) (i) Amina had $2\frac{1}{2}$ litres of water and drank $\frac{1}{3}$ litres of it. How many litres of water is she left with?
(ii) Find 0.893×14.7, correct to 4 decimal places.

(b) The mean height of 28 students in a class is 162 cm. A new girl, whose height is 149 cm, joins the class. What is the new mean height of the class now?

(c) (i) Find x if $(-3)^{x-2} = -243$

(ii) The water consumption pattern of a community is given by 11^{4x-3}. If x is in months, find correct to 4 significant figures, the total consumption after 2 months.

CamScanner

(d) The table below shows the number of marbles students sent to class for a Mathematics lesson.

Number of marbles (x)	Number of students (f)	f (x)
1	4	-
2	5	-
3	-	42
4	9	-
5	-	30
6	2	12
	$\sum f =$	$\sum f x =$

(i) Copy and complete the table
(ii) How many students were in the class?
(iii) How many marbles were brought altogether?
(iv) What is the highest number of marbles brought by a student?

3. (a) In a day, Mark earns Gh¢ 8.25 per hour working for 8 hours. He also earns overtime pay, which is $1\frac{1}{2}$ times his regular rate of Gh¢ 8.25 for 4 hours. How much pay does he earn altogether in a day?

(b) Triangle QPR below is a right – angled triangle.



Find
(i) |PQ|
(ii) Tan θ

Acadmicprep.com
Learn • Practice • Excel

(c) Find the truth set of $6 \geq 2 - 3x$ and represent your result on the number line.

(d) Find the area of a semi–circle whose radius is 7cm. [Take $\pi = \frac{22}{7}$]

4. (a) The numbers of pupils in a primary school is given in the table below.

CLASS	One	Two	Three	Four	Five	Six
NUMBER OF PUPILS	35	45	24	33	23	20

(i) Draw a pie chart for the data
(ii) What fraction of the pupils are in primary two?
(iii) Calculate the percentage of pupils in primary three, four and five altogether, correct to 2 decimal places.

(b) A straight line has a gradient of 7 and passes through the point: P (- 2, - 3). Find
(i) the equation of the line;
(ii) the value of k if the point R (k, 4) lies on the line.

Turn over

CamScanner

(c) Evaluate $\frac{2}{3} \times \frac{1}{4} + \frac{2}{5} \div \frac{14}{15}$

(d) The distance from village **A** to village **B** is $(x + 2)$ km and the distance from village **C** to village **D** is $(2x + 1)$. If these distances represent the length and breadth of a rectangular route around the villages, and the total perimeter of the route is 48 km, calculate the value of x.

5. (a) In a pen bag, 5 are red, 3 are blue and 2 are green.

(i) What is the probability of picking a red pen?

(ii) Express the probability of picking a blue pen as a

(α) decimal;

(β) percentage;

(γ) ratio.

(iii) If 2 pens are picked in succession without replacement, what is the probability that both are green?

(b) Asamoah left Takoradi at 11:13am and got to Tema at 3:08pm. How long did the journey take?

(c) If a water pump delivers 20 gallons of water per minute and Kofi can only afford half an hour cost of pumping water to a polytank, how many gallons of water will be pumped into the tank?

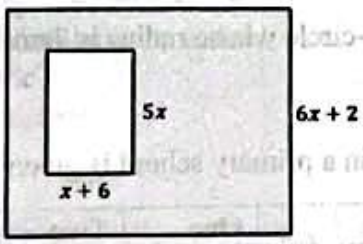
(d) In a market of 36 fishmongers, 28 sell mackerel lessons, 18 sell tilapia and 13 of them sell both.

(i) Represent this information on a Venn diagram.

(ii) How many in the market are not selling any of the two fishes?

Academicprep.com
Learn • Practice • Excel

6. (a) Find the area of the shaded region in the figure below.



(b) Solve $\frac{4}{3x} = 7 - \frac{3}{x}$

(c) Simplify $\frac{25 \times t^{-4}}{5^{-3} \times 10 \times t^{-8}}$

(d) $a = \begin{pmatrix} -4 \\ p \end{pmatrix}$ and $b = \begin{pmatrix} q \\ 1 \end{pmatrix}$. Given $a + b = \begin{pmatrix} -1 \\ -8 \end{pmatrix}$, find the value of p and q.

END OF ESSAY TEST

CamScanner

Answer **all** *questions*
Each question is followed by **four** *options* **A** *to* **D**. *Find out the* **correct** *option for* **each** *question and shade* **in pencil** *on your answer sheet, the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An example is given below.*

What is the smallest number which is divisible by 16 and 20?
A. 80
B. 40
C. 120
D. 160

The correct answer is 80, *which is lettered A and therefore answer space* **A** *would be shaded.*

[■] [B] [C] [D]

Think carefully before you shade the answer spaces. Erase completely an answer you wish to change.
Do all rough work on this question paper.
Now answer the following question.

1. Find the value of 124.3 + 0.275 + 74.06, correct to one decimal place.
   A. 198.6
   B. 198.7
   C. 892.0
   D. 892.4
2. Solve the inequality: $2x + 1 > 9$.
   A. $x > 3$
   B. $x > 4$
   C. $x > 2$
   D. $x > 1$
3. A football match starts at 1:20 pm and lasts for 1 hour 50 minutes. At what time will the match end?
   A. 4:10 pm
   B. 5:10 pm
   C. 3:10 pm
   D. 6:10 pm
4. The sum of three consecutive odd integers is 33. Find the numbers.
   A. 9, 11 and 13
   B. 7, 9 and 11
   C. 11, 13 and 15
   D. 13, 15 and 17
5. If the area of a circle is 64π, what is the circumference of the circle?
   A. 8π
   B. 16π
   C. 32π
   D. 64π
6. 9 men can complete a piece of work in 51 days. In how many days will 17 men of the same strength finish the same work?
   A. 29 days
   B. 31 days
   C. 27 days
   D. 37 days
7. The mean of the numbers 5, 8, 4, 7 and $2x$ is 6. Find x
   A. 6
   B. 4
   C. 3
   D. 2
8. Kojo's age in the next 10 years will be 4 times his age 5 years ago. How old is Kojo now?
   A. 5
   B. 6
   C. 10
   D. 15
9. Which of the following is not equivalent to $3 \times 3 \times 3 \times 3 \times 3 \times 3$?
   A. 3^6
   B. 18
   C. [illegible]
   D. 729
10. Find the value of $12x + 9q$ when $x = 11$ and $q = 3$
    A. 148
    B. 159
    C. 168
    D. 172
11. What is the locus of points that are a fixed distance from a given point?
    A. A straight line
    B. A circle
    C. An arc
    D. A rectangle
12. If $\frac{n}{100} = 10.5$, find the value of n.
    A. 0.0105
    B. 0.105
    C. 1050
    D. 105

Academicprep.com
Learn • Practice • Excel

CamScanner

13. A is the point with coordinates (2,3). If $\overrightarrow{AB} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}$, Find the coordinates of B.
A. (7, -1)
B. (-3, 7)
C. (3, -7)
D. (5, 4)

14. Evaluate: $(0.07 \times 0.02) \div 14$
A. 0.01
B. 0.001
C. 0.0001
D. 0.011

15. Which of the following sets is the largest?
A. {whole numbers}
B. {composite numbers}
C. {natural numbers}
D. {integers}

16. Sam has $5xy$ toffees and Joe has $20xy$ toffees. How many more toffees has Joe?
A. $15xy$
B. $15x$
C. $15y$
D. $4xy$

17. Which of the following could be used to find x in the diagram below?



A. Add 2.1 and 3.4, then subtract the sum from 9.7
B. Add 9.7 and 3.4, then subtract 2.1 from the sum
C. Subtract 2.1 from 3.4, then subtract the difference from 9.7
D. Subtract 3.4 from 9.7, then add 2.1 to the difference

18. Find the solution set of the inequality $5 - 5(3 - y) \leq 65$, if y = {one of the first 6 multiples of 5}
A. {y:y = 5, 10, 15}
B. {y:y = 5, 10}
C. {y:y = 5,10, 15, 20}
D. {y:y = 5, 10,15, 20, 25, 30}

19. Mina is x years old now. How old will she be in $10 + y$ years?
A. $(x - 10 + y)$ years
B. $(x - 10 - y)$ years
C. $(x + 10 + y)$ years
D. $(x + 10 - y)$ years

20. A line passes through (1, 2) and (4, y). what is the value of y if the gradient is -2?
A. 4
B. - 2
C. 0
D. - 4

21. If $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} -4 \\ 5 \end{pmatrix}$, find $\overrightarrow{BC}$.
A. $\begin{pmatrix} 7 \\ 10 \end{pmatrix}$
B. $\begin{pmatrix} -7 \\ -10 \end{pmatrix}$
C. $\begin{pmatrix} 7 \\ -10 \end{pmatrix}$
D. $\begin{pmatrix} -7 \\ 0 \end{pmatrix}$

22. A line passes through points (2, 6) and (4, 12). Find the constant of proportionalit
A. 1
B. 2
C. 3
D. 6

23. A van travels 154km in $1\frac{3}{4}$ hours. Find its speed in km/ h.
A. 77 km/h
B. 88 km/h
C. 100 km/h
D. 269 km/h

24. A hall which is 20m long is represented on a diagram as 10cm long. What is the scale of the diagram?
A. 1:200
B. 1:250
C. 1:400
D. 1:500

25. If $13x - 12 = 5x + 60$, find x.
A. – 9
B. – 6
C. 9
D. 6

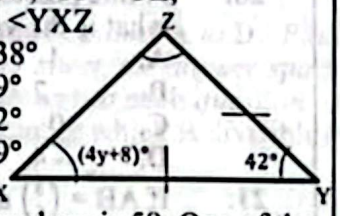
26. Order the fractions $\frac{3}{8}, \frac{1}{2}, \frac{5}{16}$ from the smallest to the largest.
A. $\frac{3}{8} < \frac{1}{2} < \frac{5}{16}$
B. $\frac{1}{2} < \frac{3}{8} < \frac{5}{16}$
C. $\frac{5}{16} < \frac{3}{8} < \frac{1}{2}$
D. $\frac{1}{2} < \frac{5}{16} < \frac{3}{8}$

Acadmicprep.com
Learn • Practice • Excel

Turn over

CamScanner

27. In the triangle below, XY = YZ, <XYZ = 42°. Find <YXZ



A. <YXZ = 138°
B. <YXZ = 59°
C. <YXZ = 42°
D. <YXZ = 69°

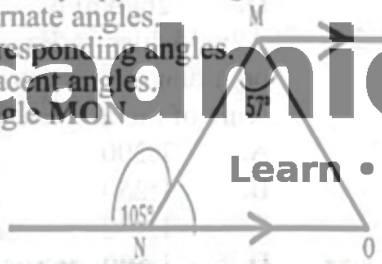
28. The mean of ten numbers is 58. One of the numbers is 40. What is the mean of the other nine?
A. 54
B. 56
C. 58
D. 60

29. The population of pigs in a certain farm is 5 $(2x - 3)$, where x is in months, find the population of the pigs in the farm after 4 months.
A. 52
B. 25
C. 30
D. 15

Use the diagram below to answer questions **30** *and* **31**

30. Angle MON and angle PMO are
A. vertically opposite angles.
B. alternate angles.
C. corresponding angles.
D. adjacent angles.

31. Find the angle MON



A. 48°
B. 52°
C. 58°
D. 65°

32. A fly weighs 0.1kg. How much would 112 flies weigh combined?
A. 1.12 kg
B. 112 kg
C. 11.2 kg
D. 0.112 kg

33. The population growth of a given town is given by 2^{2m} where m is in years. Find the time needed for the population to be 4^{14}.
A. 7
B. 21
C. 14
D. 28

34. Which of the following is an example of numerical data?
A. Eye color
B. Height in metres
C. Type of car
D. Favourite food

The graph below represents the heights a walker reached during a long trail.

Use it to answer questions **35** *to* **37.**



35. Which section of the graph shows the steepest positive gradient?
A. Section A
B. Section B
C. Section H
D. Section F

36. Which section of the graph shows the shallowest positive gradient?
A. Section A
B. Section B
C. Section F
D. Section G

37. In which section of the graph is the gradient zero?
A. Section A
B. Section B
C. Section D
D. Section E

38. Given that $P = \begin{pmatrix} 3x-6 \\ 2y+5 \end{pmatrix}$, $R = \begin{pmatrix} 9 \\ 7 \end{pmatrix}$ and P = R, find the values of x and y.
A. $x = 1$ and $y = 5$
B. $x = -5$ and $y = -1$
C. $x = 5$ and $y = 1$
D. $x = -1$ and $y = 5$

39. Solve for x in the equation $15 - 2x = 6$
A. − 10.5
B. − 4.5
C. 4.5
D. 10.5

40. A baker has 12 identical cakes. In how many different ways can he divide them up into equal packets?
A. 4
B. 5
C. 6
D. 7

END OF PAPER

Academicpro.com
Learn • Practice • Excel

CamScanner